

Initial Project Planning Template

Date	06 July 2026
Project Name	Credit Card Approval Prediction System
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Product Backlog, Sprint Schedule, and Estimation

Backlog and sprint plan for building the Credit Card Approval Prediction system, from data preparation and model training through to the Flask web app and deployment.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Collection & Preprocessing	USN-1	As a developer, I can load and explore application_record.csv and credit_record.csv so I understand the dataset structure and fields.	1	High	Joyal Kumar	15 Mar 2026	29 Mar 2026
Sprint-1		USN-2	As a developer, I can remove duplicate applicants and handle missing OCCUPATION_TYPE values so the dataset is clean for modeling.	2	High	Keerthi	15 Mar 2026	29 Mar 2026
Sprint-1		USN-3	As a developer, I can convert the monthly credit STATUS into a single Good/Bad target per customer so each applicant has one final label.	2	High	Yasawini	15 Mar 2026	29 Mar 2026
Sprint-1	Feature Engineering	USN-4	As a developer, I can label-encode categorical fields (gender, education, income type, etc.) and merge application and credit data so it's ready for training.	3	High	Nasir	15 Mar 2026	29 Mar 2026
Sprint-1	Model Building	USN-5	As a developer, I can train and compare Logistic Regression, Decision Tree, and Random Forest models so I can identify the best-performing classifier.	3	High	Harika	15 Mar 2026	29 Mar 2026
Sprint-1		USN-6	As a developer, I can view accuracy, F1 score, and a confusion matrix for each model so I can justify the final model choice.	2	Medium	Joyal Kumar	15 Mar 2026	29 Mar 2026
Sprint-1		USN-7	As a developer, I can save the best-performing model as model.pkl so it can be reloaded for predictions without retraining.	1	Medium	Keerthi	15 Mar 2026	29 Mar 2026
Sprint-2	Web Application - Frontend	USN-8	As an applicant, I can fill out a form with my personal and financial details so I can check my credit approval chances.	2	High	Yasawini	30 Mar 2026	13 Apr 2026
Sprint-2	Web Application - Backend	USN-9	As an applicant, I can submit the form and receive a Good/Bad Customer prediction so I know my likely approval outcome.	3	High	Nasir	30 Mar 2026	13 Apr 2026
Sprint-2		USN-10	As an applicant, I can view my prediction result on a clear results page so the outcome is easy to understand.	1	Medium	Harika	30 Mar 2026	13 Apr 2026

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Sprint-2		USN-11	As an applicant, I receive a clear error message if I submit invalid or incomplete data so I can correct my entry.	2	Medium	Joyal Kumar	30 Mar 2026	13 Apr 2026
Sprint-2		USN-12	As a developer, I can keep the label encoder consistent between training and the live app so predictions aren't affected by mismatched encodings.	2	High	Keerthi	30 Mar 2026	13 Apr 2026
Sprint-3	Deployment & Documentation	USN-13	As a developer, I can add a requirements.txt and .gitignore so others can set up and run the project easily.	1	Low	Yasawini	14 Apr 2026	28 Apr 2026
Sprint-3		USN-14	As a developer, I can deploy the Flask app to a hosting platform so users can access it via a public URL instead of running it locally.	3	Medium	Nasir	14 Apr 2026	28 Apr 2026
Sprint-3	Testing	USN-15	As a developer, I can test the form with edge-case and invalid inputs so the app fails gracefully instead of crashing.	2	Medium	Harika	14 Apr 2026	28 Apr 2026